Group : Group 13

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Derivation of the gradient

$J = (1/N) \cdot e^T e = (1/N) \cdot (xm + b - y)^T (xm + b - y)$

$\partial e / \partial m = x \qquad \partial e / \partial b = I$

$\partial (e^T e) / \partial e = 2e$

$\partial J / \partial m = (1/N) \cdot x^T \cdot 2e \rightarrow \partial J / \partial m = (2/N)\, x^T (\hat{y} - y)$

$\partial J / \partial b = (1/N) \cdot I^T \cdot 2e \rightarrow \partial J / \partial b = (2/N)(\hat{y} - y)$

$m_new = m - a \cdot \partial J / \partial m$

$b_new = b - a \cdot \partial J / \partial b$

Since $N = 2$, factor $2/N = 1$

$x^T \begin{vmatrix} 1 & 4 \\ 3 & 10 \end{vmatrix}$

Update 1 (Gatete Derrick)

$m = [-1, 2]^T \qquad b = [1, 1]^T$

$\hat{y} = x \cdot m + b$

$= \begin{vmatrix} 1(-1) + 3(2) \\ 4(-1) + 10(2) \end{vmatrix} + \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \begin{vmatrix} 5 \\ 16 \end{vmatrix} + \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \begin{vmatrix} 6 \\ 17 \end{vmatrix}$

$e = \hat{y} - y = \begin{vmatrix} 6-5 \\ 17-6 \end{vmatrix} = \begin{vmatrix} 1 \\ 11 \end{vmatrix}$

$J = (1^2 + 11^2)/2 = 61$

$\partial J / \partial m = x^T e \begin{vmatrix} 1(1) + 4(11) \\ 3(1) + 10(11) \end{vmatrix} = \begin{vmatrix} 45 \\ 113 \end{vmatrix}$

$\partial J / \partial b = e = [1, 11]^T$

$m_new = \begin{vmatrix} -1 - 0.01(45) \\ 2 - 0.01(113) \end{vmatrix} = \begin{vmatrix} -1.4500 \\ 0.8700 \end{vmatrix}$

$b_new = \begin{vmatrix} 1 - 0.01(1) \\ 1 - 0.01(11) \end{vmatrix} = \begin{vmatrix} 0.9900 \\ 0.8900 \end{vmatrix}$

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Update 2 (Igor Ntwari)

$m = [-1.45, 0.87]^T \quad b = [0.99, 0.89]^T$

$$\hat{y} = \begin{vmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{vmatrix} + \begin{vmatrix} 0.99 \\ 0.89 \end{vmatrix} = \begin{vmatrix} 1.16 \\ 2.90 \end{vmatrix} + \begin{vmatrix} 0.99 \\ 0.89 \end{vmatrix} = \begin{vmatrix} 2.15 \\ 3.79 \end{vmatrix}$$

$$e = \begin{vmatrix} 2.15 - 5 \\ 3.79 - 6 \end{vmatrix} = \begin{vmatrix} -2.85 \\ -2.21 \end{vmatrix}$$

$J = (8.1225 + 4.8841)/2 = 6.5033$

$$\partial J / \partial m = x^T e = \begin{vmatrix} 1(-2.85) + 4(-2.21) \\ 3(-2.85) + 10(-2.21) \end{vmatrix} = \begin{vmatrix} -11.69 \\ -30.65 \end{vmatrix}$$

$\partial J / \partial b = e = [-2.85, -2.21]^T$

$$m\text{-new} = \begin{vmatrix} -1.45 - 0.01(-11.69) \\ 0.87 - 0.01(-30.65) \end{vmatrix} = \begin{vmatrix} -1.3331 \\ 1.1765 \end{vmatrix}$$

$$b\text{-new} \begin{vmatrix} 0.99 - 0.01(-2.85) \\ 0.89 - 0.01(-2.21) \end{vmatrix} = \begin{vmatrix} 1.0185 \\ 0.9121 \end{vmatrix}$$

Update 3 (Alieu Jobe)

$m = [-1.3331, 1.1765]^T \quad b = [1.0185, 0.9121]^T$

$$\hat{y} = \begin{vmatrix} 1(-1.3331) + 3(1.1765) \\ 4(-1.3331) + 10(1.1765) \end{vmatrix} + \begin{vmatrix} 1.0185 \\ 0.9121 \end{vmatrix} = \begin{vmatrix} 2.1964 \\ 6.4326 \end{vmatrix} + \begin{vmatrix} 1.0185 \\ 0.9121 \end{vmatrix} = \begin{vmatrix} 3.2149 \\ 7.3447 \end{vmatrix}$$

$$e = \begin{vmatrix} 3.2149 - 5 \\ 7.3447 - 6 \end{vmatrix} = \begin{vmatrix} -1.7851 \\ 1.3447 \end{vmatrix}$$

$J = (3.1866 + 1.8082)/2 = 2.4974$

$$\partial J / \partial m = x^T e = \begin{vmatrix} 1(-1.7851) + 4(1.3447) \\ 3(-1.7851) + 10(1.3447) \end{vmatrix} = \begin{vmatrix} 3.5937 \\ 8.0917 \end{vmatrix}$$

$\partial J / \partial b = e = [-1.7851, 1.3447]^T$

$$m\text{-new} = \begin{vmatrix} -1.3331 - 0.01(3.5937) \\ 1.1765 - 0.01(8.0917) \end{vmatrix} = \begin{vmatrix} -1.3690 \\ 1.0956 \end{vmatrix}$$

$$b\text{-new} = \begin{vmatrix} 1.0185 - 0.01(-1.7851) \\ 0.9121 - 0.01(1.3447) \end{vmatrix} = \begin{vmatrix} 1.0364 \\ 0.8987 \end{vmatrix}$$

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Summary

- After each update, m and b move in the direction opposite to the gradient, and the cost J fall from 61 to 6.5033 to 2.4974 to 2.1645. So yes, the parameters are moving in a direction that reduces the error. The first step is large because the initial error was large. Update 2 overshot slightly ($\hat{y}$ went below y), so the gradient changed sign and pulled the parameters back. The steps then get smaller as the cost approaches its minimum, which shows ~~love~~ convergence.

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